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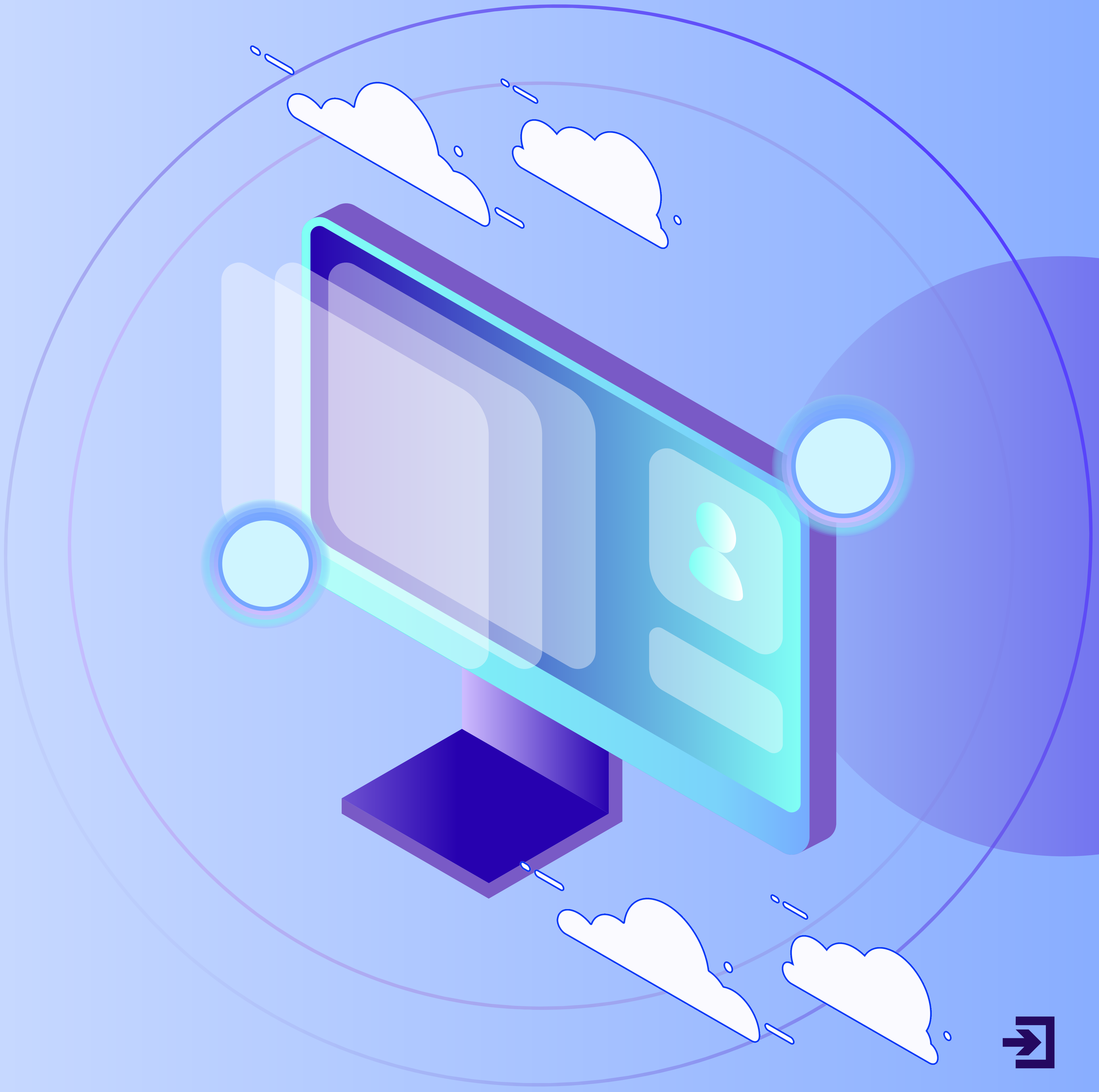
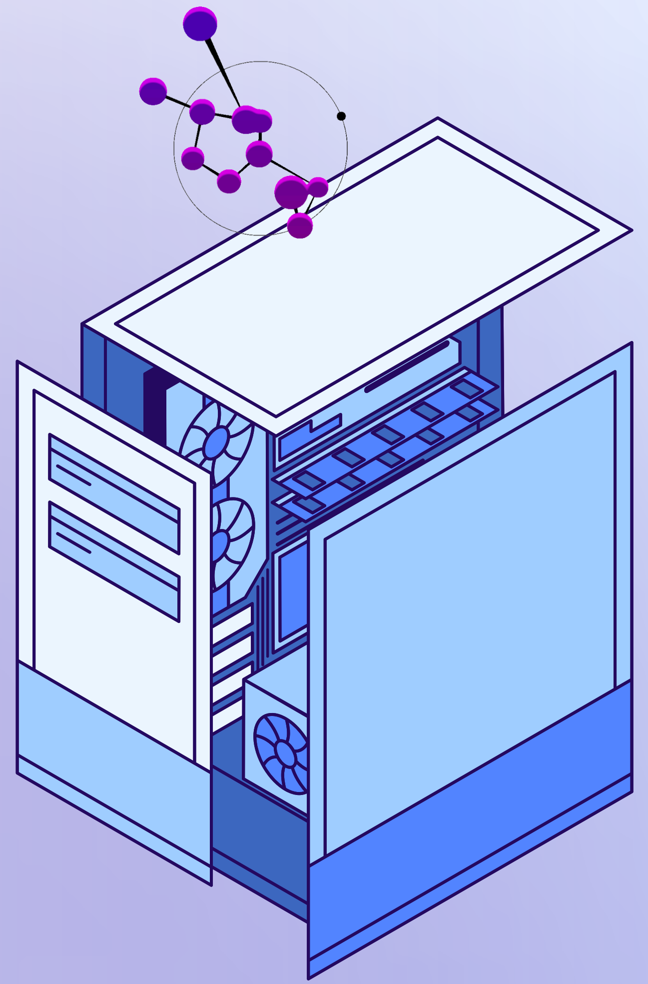
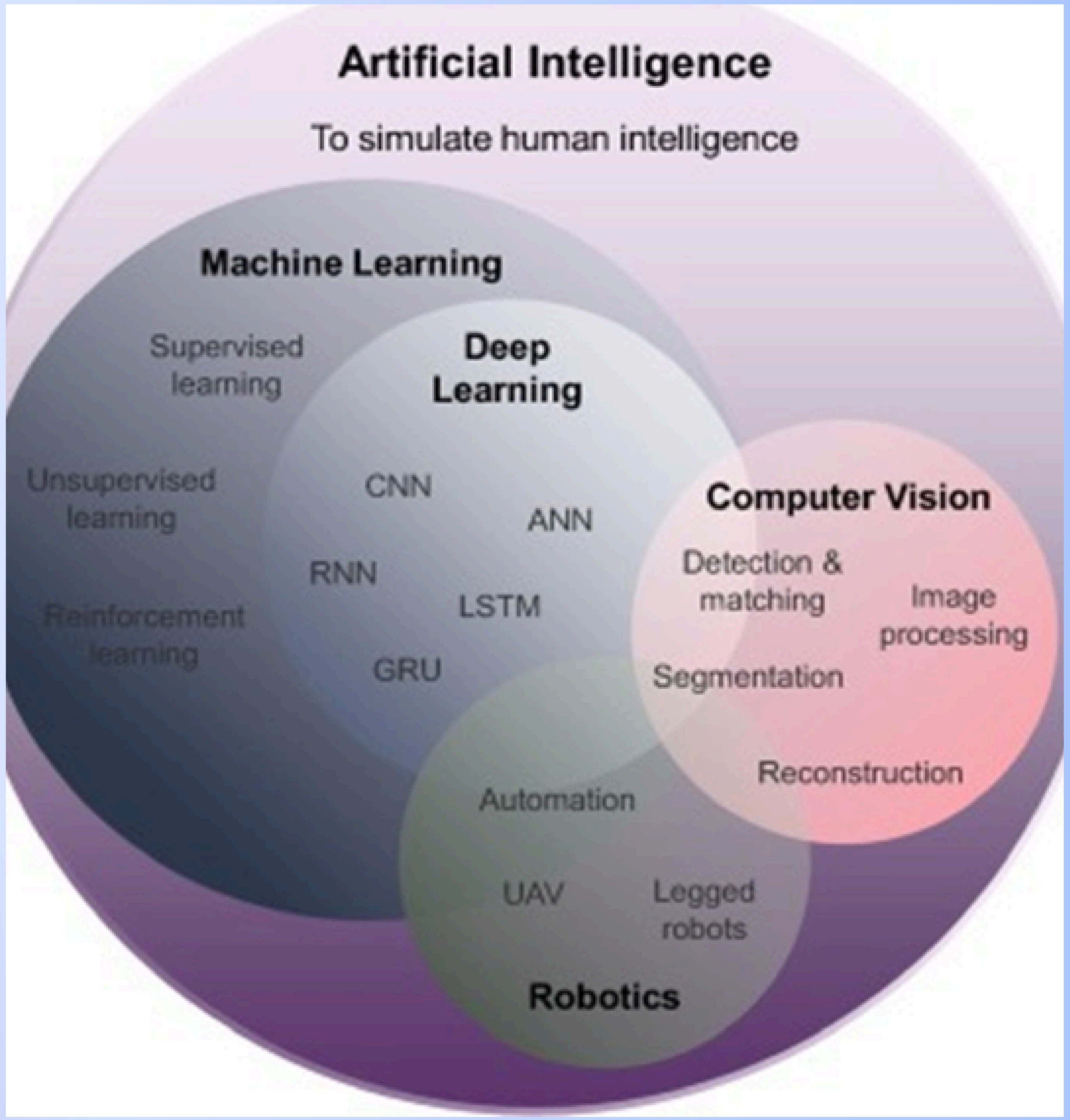
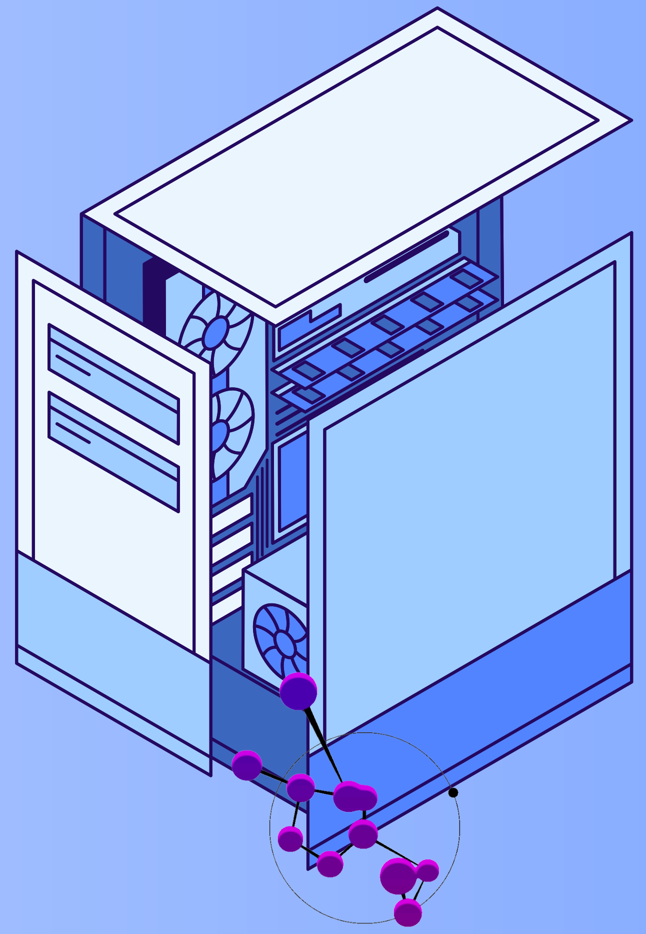
COMPUTER VISION






Artificial Intelligence
To simulate human intelligence
Machine Learning
Supervised learning
Unsupervised learning
Reinforcement learning
Deep Learning
CNN
ANN
RNN
LSTM
GRU
Computer Vision
Detection & matching
Image processing
Segmentation
Reconstruction
Automation
UAV
Legged robots
Robotics



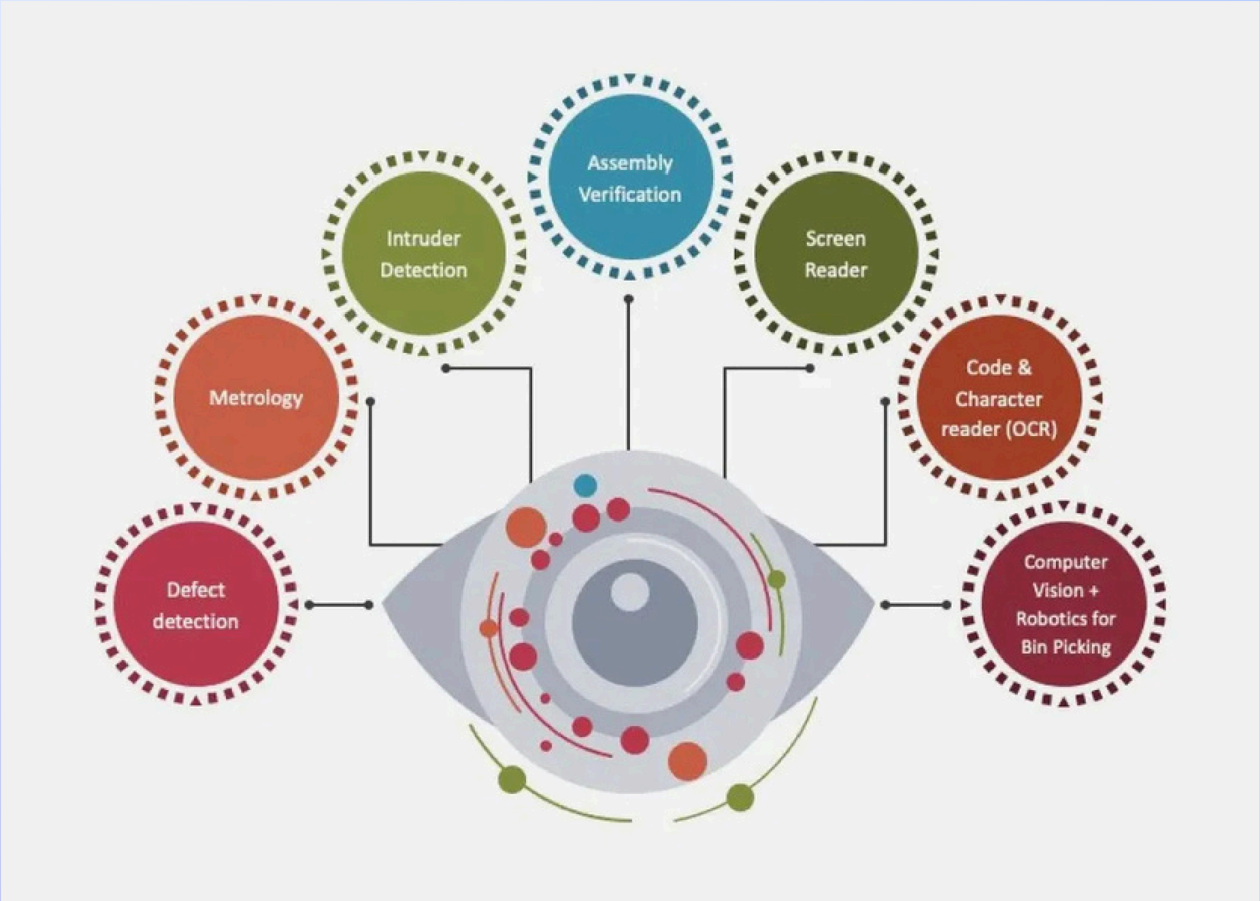
Computer Vision is a branch of Artificial Intelligence (AI) that enables computers to interpret and extract information from images and videos, similar to human perception. It involves developing algorithms to process visual data and derive meaningful insights.

Why Learn Computer Vision?

1)High Demand in the Job Market: Essential for careers in AI, machine learning, and data science across industries like healthcare, automotive, and robotics.

2)Revolutionizing Industries: Powers advancements in self-driving cars, medical diagnostics, agriculture, and manufacturing by automating visual tasks.

3)Solving Real-World Problems: Enhances public safety, improves medical imaging, and optimizes industrial processes.



This Computer Vision tutorial is designed for both beginners and experienced professionals, covering key concepts of computer vision, including Image Processing, Feature Extraction, Object Detection and Recognition, and Image Segmentation.

KEY CONCEPTS

1.Introduction
2.Image Acquisition
3.Image Preprocessing
4.Object Detection or Feature Extraction
5.Image Segmentation
6.Classification / Recognition
7.Post-processing
8.Decision-Making / Deployment
9.Machine Learning for Computer Vision
10.Deep Learning in Computer Vision
11.Real-World Use Case: Smart Agriculture

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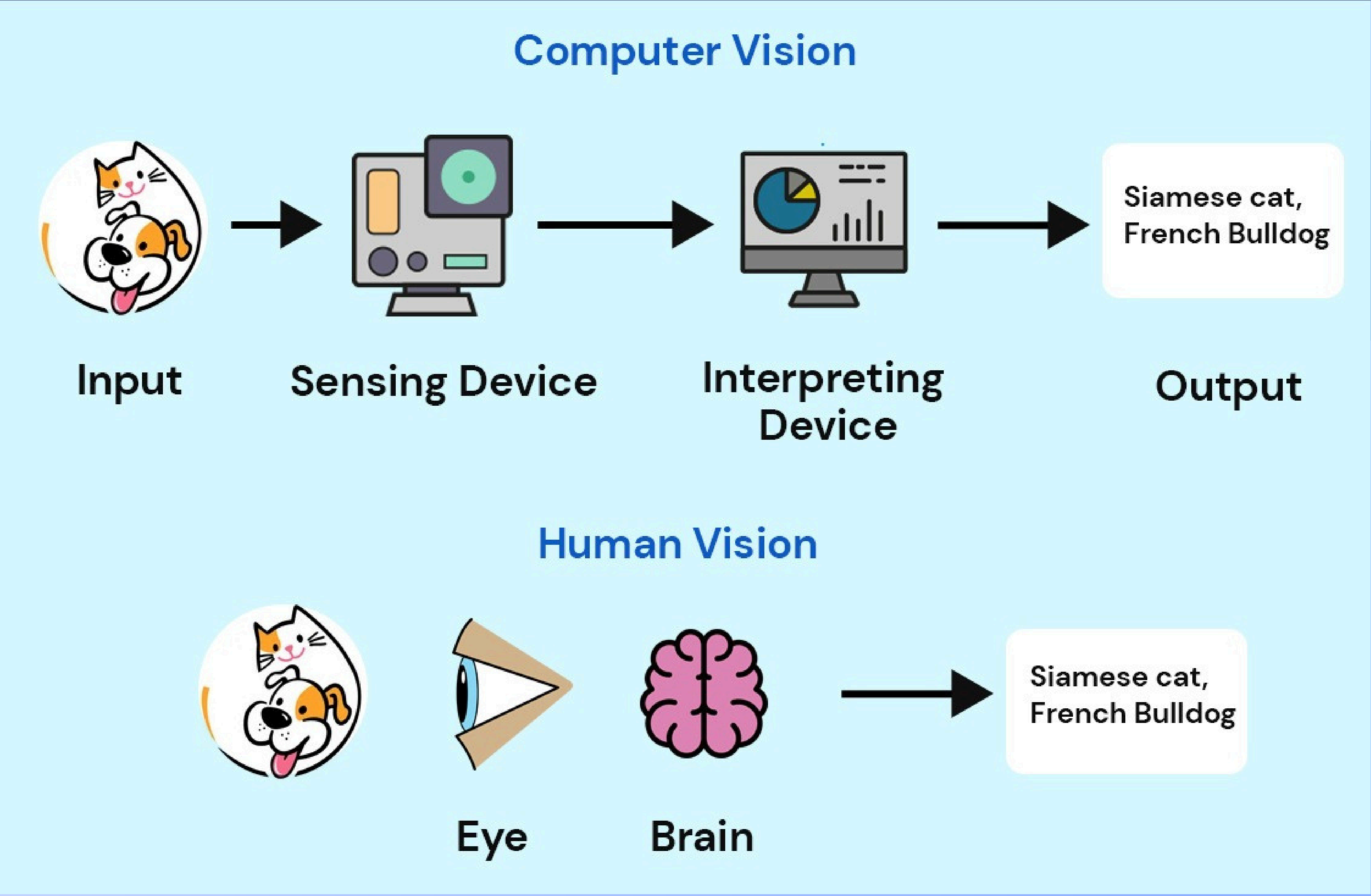
How Does Computer Vision Work?



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INTRODUCTION TO COMPUTER VISION

Computer Vision is a field of Artificial Intelligence (AI) that enables machines to interpret and understand visual data (images, videos).

Example:
1.A self-driving car uses CV to detect pedestrians, traffic signs, and other vehicles.
2.Facebook uses facial recognition to tag people in photos.

• Image Acquisition

This is the first step in the pipeline. The system gathers visual data from the environment using devices like:

*Digital cameras
*Webcams
*Smartphones
*Surveillance systems
*Drones
*Medical imaging devices (X-ray, MRI)

🔹 Data Types: Static image (JPG, PNG) or video (MP4, AVI)
📌 Example: A drone flying over farmland captures images of crops.

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IMAGE PREPROCESSING

This step improves image quality and prepares it for analysis. This is critical because raw images can be noisy, too large, or inconsistent.

🧰 **Techniques:**

- Resizing: Standardize all images (e.g., 224x224 pixels for CNN input)
- Grayscale Conversion: Reduces complexity (RGB → Single channel)
- Histogram Equalization: Enhances contrast
- Denoising: Reduces image noise using filters (Gaussian blur)
- Normalization: Pixel values scaled (0 to 1 or -1 to 1)

📌 **Example**: Converting a 4K image to 256x256 grayscale before classification.

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OBJECT DETECTION OR FEATUREXTRACTION

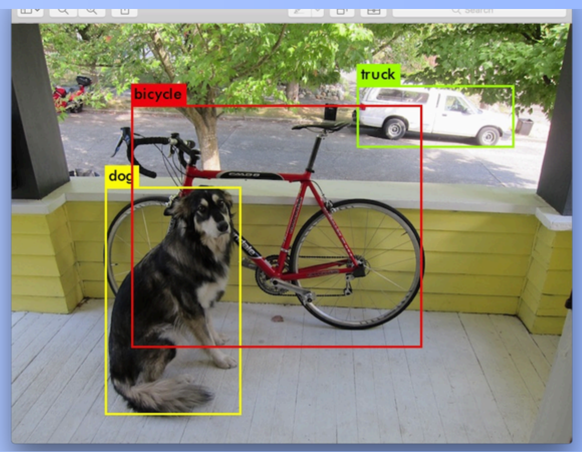
In this step, important objects or features are identified from the image.

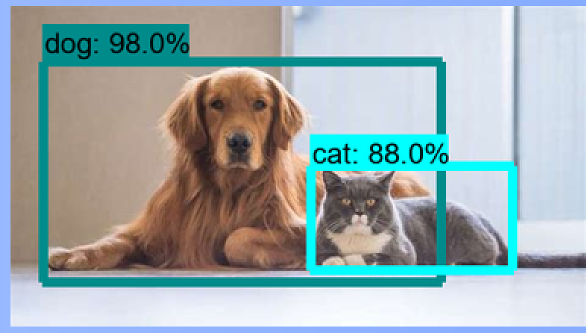
🔍 **Feature Extraction Techniques:**

- Traditional: SIFT, SURF, ORB (detect key points)
- Modern: Deep learning using Convolutional Neural Networks (CNN)

📦 **Object Detection Models:**

- YOLO (You Only Look Once)
- Faster R-CNN
- SSD (Single Shot Detector)





📌 **Example:** Detect faces, pedestrians, or number plates in a traffic camera feed.

IMAGE SEGMENTATION

This step divides the image into meaningful regions, helping the model to focus on objects.

🧪 **Types:**

- Semantic Segmentation: Classifies every pixel (e.g., road, building, sky)
- Instance Segmentation: Identifies individual objects (e.g., car1, car2)

📌 **Example**: In medical imaging, identify tumor boundaries in an MRI scan.

🔍 HOW YOLO WORKS

1.Single Neural Network processes the entire image at once.
2.It divides the image into a grid, and each grid cell predicts:

- Bounding boxes (object position)
- Class probabilities (what the object is)
- Confidence scores (how likely it's correct)

3.Instead of scanning parts of the image multiple times (like older methods), YOLO looks at the entire image once, making it extremely fast.

✅ ***Advantages of YOLO:***

***Speed: Great for real-time applications (e.g., surveillance, autonomous driving).**
***Global context: Since it sees the whole image at once, it reduces false positives.**

🚀 ***Variants:***

- **YOLOv1 → First version**
- **YOLOv2 / YOLO9000 → Improved speed & accuracy**
- **YOLOv3 → More accurate on smaller objects**
- **YOLOv4 & v5 → Faster and lightweight (v5 is PyTorch-based)**
- **YOLOv7, YOLOv8 → Even better for edge devices, highly optimized**

🔧 ***Use Cases:***

- **Self-driving cars**
- **Face detection**
- **Retail analytics**
- **Security surveillance**

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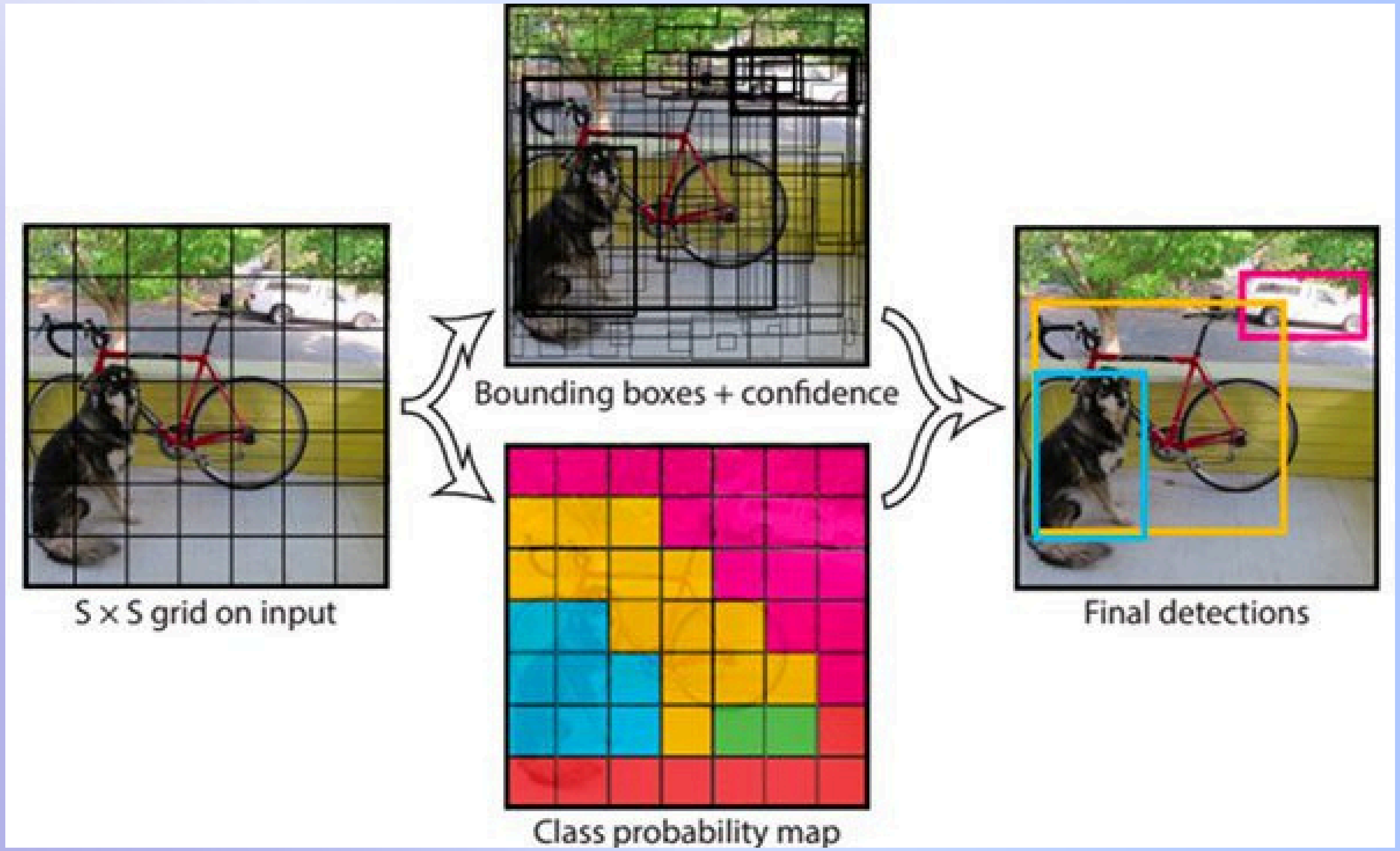
YOLO object detection with OpenCV



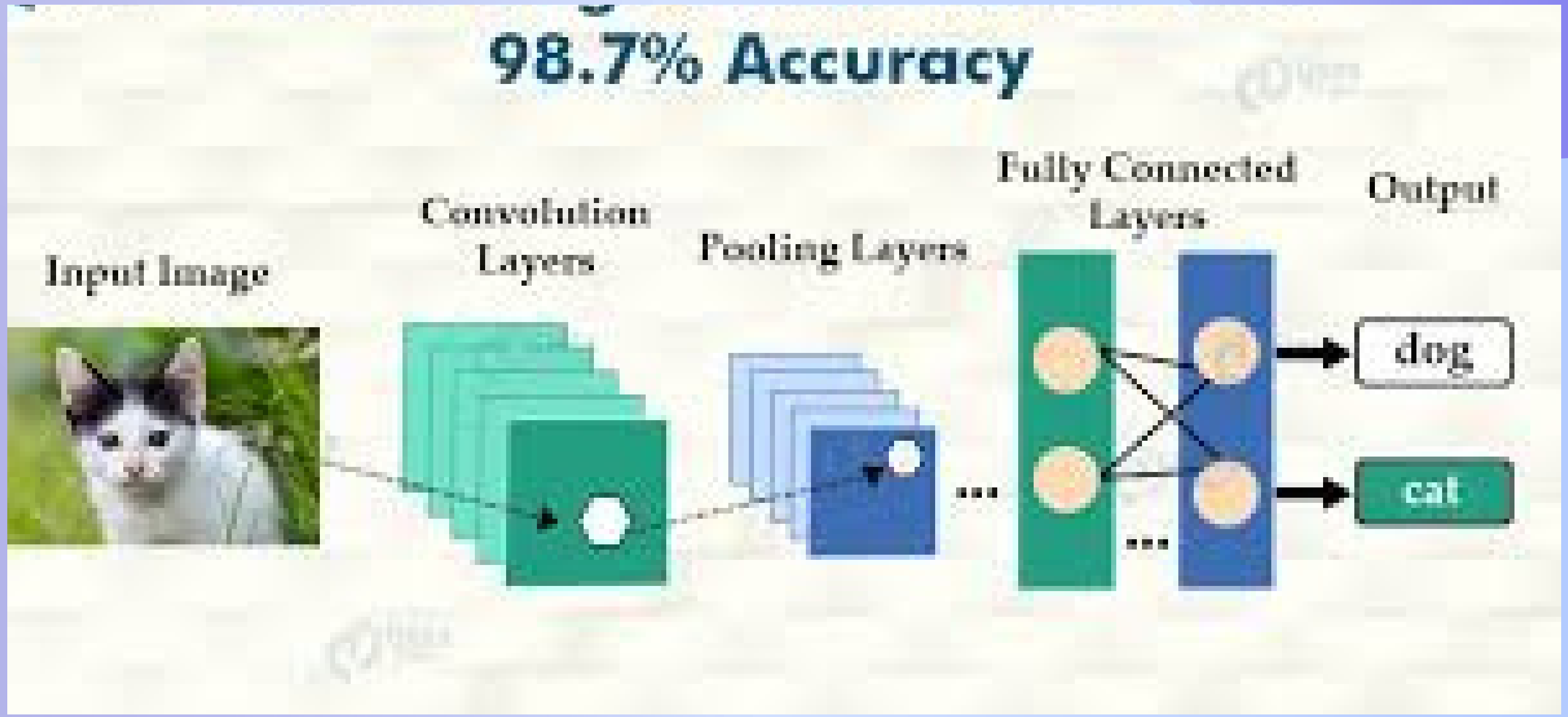
CLASSIFICATION / RECOGNITION

What is Image Classification?
Definition: Assigning a **single label** to an entire image based on its dominant content.
Example: Cat vs. Dog Classifier
Model Output:

- **Class: "Cat"**
- **Confidence: 98%**

How It Works:

1. **A CNN (e.g., ResNet) extracts features (edges, whiskers, fur patterns).**
2. **The final softmax layer assigns probabilities to predefined classes.**

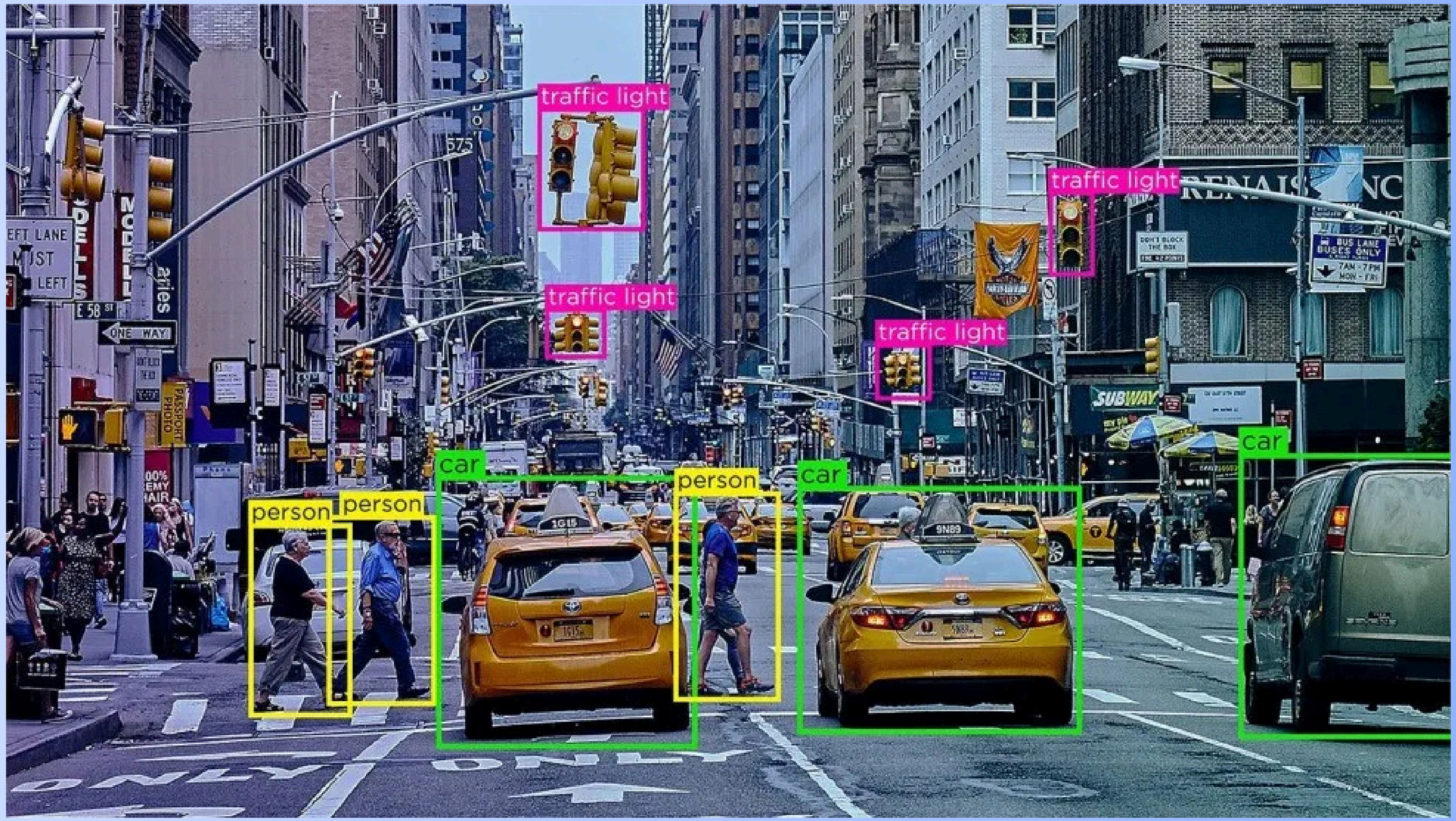


Recognition

What is Object Recognition?
Detecting and classifying multiple objects within an image (with bounding boxes).
Example: YOLO Object Detection

Model Output:

Object	Confidence	Bounding Box
Car	92%	[x1, y1, x2, y2]
Pedestrian	85%	[x1, y1, x2, y2]
Traffic Light	78%	[x1, y1, x2, y2]



Key Difference:
- **Classification:** One label per image.
- **Recognition:** Multiple labels + locations.

Post-processing

This step includes refining the results for presentation, reporting, or next action.

🔧 **Includes:**

**Drawing bounding boxes or masks
**Thresholding confidence levels
**Removing duplicates (non-max suppression)
**Exporting labels/results to a database

📌 **Example**: Bounding box around a stop sign and outputting label "Stop Sign (98%)".

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Decision-Making / Deployment

Finally, the result is used for a real-world action or decision:

🚀 **Usage Examples:**

Security: Alert if an unauthorized person is detected
Retail: Update stock when shelf is empty
Healthcare: Recommend further scan if tumor is detected
Vehicles: Stop the car if obstacle detected

📌 **Example:** A smart CCTV system alerts the security team when it detects a person with a weapon.

Machine Learning for Computer Vision

(a) Supervised Learning (Classification)

Train a model (e.g., SVM, Random Forest) on labeled images.
Example: Classifying cats vs. dogs.

(b) Unsupervised Learning (Clustering)

Group similar images without labels (e.g., K-Means).
Example: Organizing photo albums by similar scenes.

Deep Learning in Computer Vision

(a) Convolutional Neural Networks (CNNs)

Specialized neural networks for image processing.

Layers: Convolution → Pooling → Fully Connected.

Example:

LeNet-5 (Digit Recognition)

AlexNet (ImageNet Classification)

(b) Transfer Learning

Using pre-trained models (e.g., ResNet, VGG16) for new tasks.

Example: Fine-tuning VGG16 for medical image diagnosis.

(c) Object Detection Models

YOLO (You Only Look Once) – Real-time detection.

Faster R-CNN – High accuracy detection.

Example: Detecting cars in traffic cameras.

(d) Semantic Segmentation

Classifies each pixel in an image (e.g., U-Net).

Example: Self-driving cars identifying roads vs. sidewalks.

Step	Task	Tools/Techniques Used	Example
Image Acquisition	Capture visual input	Camera, Drone, Smartphone	Snap photo of a leaf
Preprocessing	Enhance image quality	Resize, Denoise, Normalize	Convert to 256x256 grayscale
Object Detection	Locate important items	YOLO, R-CNN, SSD	Detect "dog" and "person"
Segmentation	Break into labeled regions	Semantic/Instance Segmentation	Highlight tumor in an X-ray
Classification	Label the object or scene	CNN, ResNet, VGG	Classify leaf as “infected”
Post-processing	Polish output and get confidence	Non-max suppression, thresholding	Show bounding box with label
Decision-Making	Act on result	Application logic or automation system	Alert farmer for pest-infected leaf

📌 Real-World Use Case: Smart Agriculture

Image Acquisition – A drone captures high-resolution images of crops.

Preprocessing – Images resized and denoised.

Object Detection – Detects damaged or diseased leaves.

Segmentation – Highlights affected areas pixel-by-pixel.

Classification – Determines disease type (e.g., leaf blight).

Post-processing – Confidence level displayed (e.g., 94% confidence).

Decision – Farmer gets a notification with image and suggestion.